



Indian Institute of Technology Guwahati

BSc. (Hons.) Data Science & Artificial Intelligence
Online Degree Programme

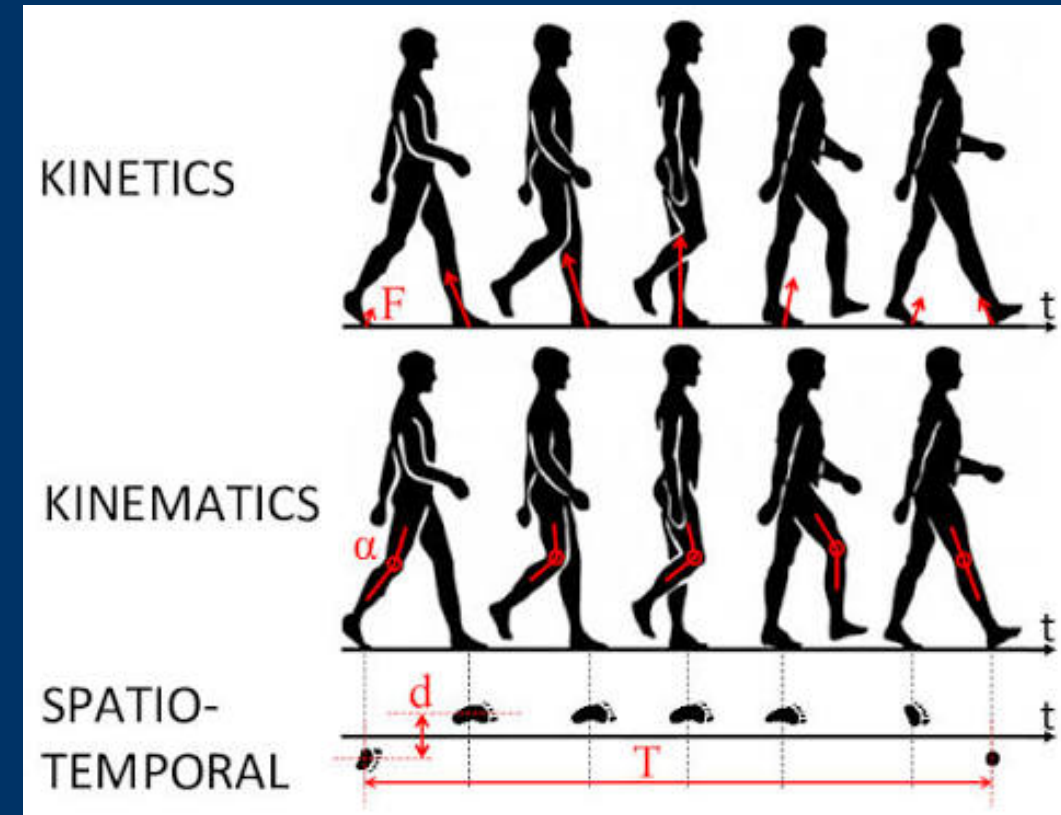
INTERNSHIP NIT SILCHAR

POST STROKE GAIT ANALYSIS BASED ON KINEMATICS DATA AND INTERPRETABILITY USING EXPLAINABLE AI TECHNIQUES

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The Problem – Why Gait Analysis After Stroke?

- Stroke survivors are frequently left with gait abnormalities that impair mobility and quality of life
- Prior work (Romano et al., 2024) showed that entropy-based complexity metrics from combined EMG + kinematics signals can classify stroke vs. healthy gait accurately
- Problem: EMG needs extra sensor placement and calibration → limits routine clinical use
- Research question: Can kinematics data alone retain the same diagnostic power? And can the model be made interpretable enough for clinicians to trust?



Our Solution – System Architecture

- 1. Data & Strict ID Tracking: Van Criekinge et al. dataset (138 healthy, 50 stroke). Joint angles extracted (Ankle, Knee, Hip, Pelvis).
- 2. Entropy Feature Engineering: Raw time-series converted into 5 complexity metrics to quantify movement irregularity.
- 3. Dynamic Class Balancing: Healthy majority (138) is iteratively downsampled to match stroke minority (50).
- 4. Leakage-Proof MRMR Selection: Feature selection runs strictly inside the inner CV loop, ensuring it never sees test data.
- 5. Multi-Model Training: 7 distinct classifiers trained using randomized hyperparameter search per fold to eliminate architectural bias.
- 6. Out-of-Fold Evaluation: Metrics (Accuracy, Precision, Recall, AUC/ROC) computed out-of-fold to expose models that merely memorized the training set.
- 7. Explainable AI (XAI): SHAP attributions aggregated per anatomical joint and cross-checked against LIME on the best model to validate reasoning.

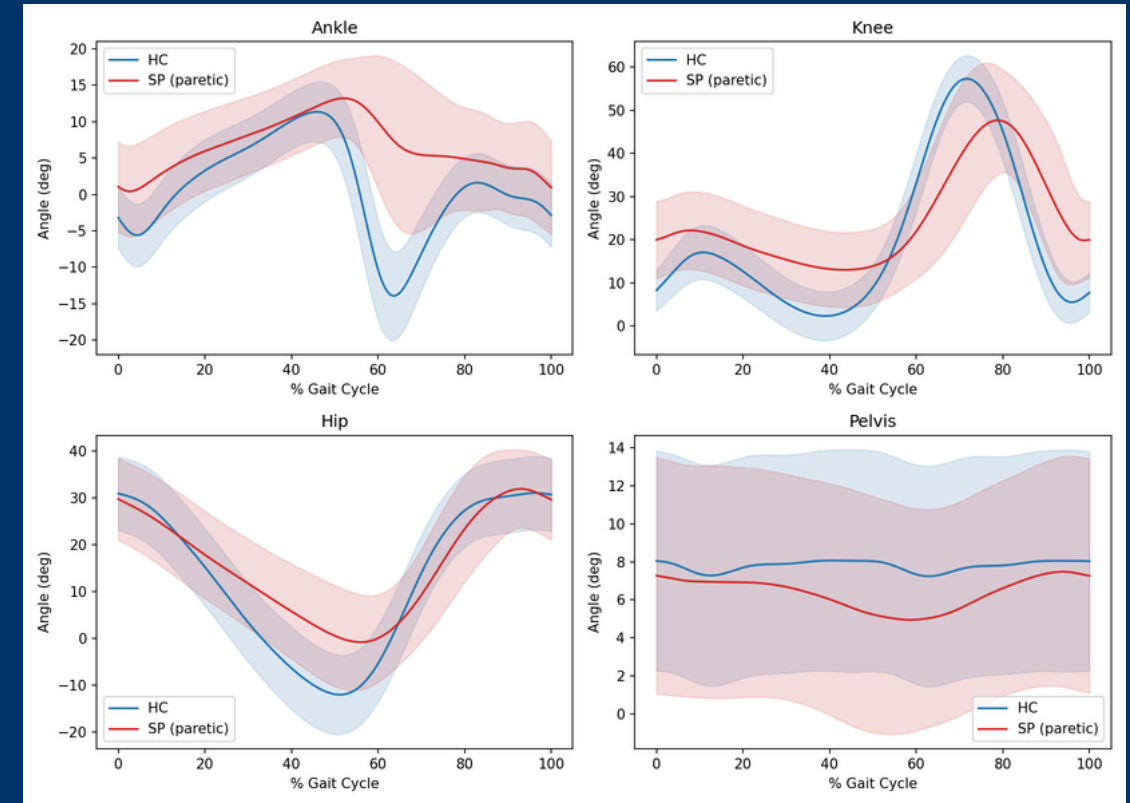


Tools & Technologies Used

- **Environment:** Python (Kaggle Notebook).
- **Data Handling:** Per-subject DataFrames/dictionaries for joint-angle time series (Pandas/NumPy).
- **Training & Validation:** 10-fold nested stratified CV with randomized hyperparameter tuning across 7 classifiers (Coarse Tree, Logistic Regression, Kernel NB, Cubic SVM, Fine KNN, Bagged Trees, Medium NN) using (Scikit-Learn)
- **Explainability:** SHAP (feature attribution aggregated per joint) cross-checked against LIME.
- **Visualization:** Trajectory plots, confusion matrices and SHAP bar/beeswarm plots using (Matplotlib)

The Dataset

- Van Criekinge et al. public kinematic gait-cycle dataset
- 138 healthy controls (HC) and 50 post-stroke individuals (SP)
- Joint angle trajectories for Ankle, Knee, Hip, Pelvis
- Right side used for HC; paretic side used for SP (per reference protocol)
- Data cleaning: subject identity preserved to prevent cross-subject leakage; quality checks run on signal length/missingness
- Class balancing: healthy group downsampled to match 50 SP → 50:50 balanced dataset per iteration



Machine Learning Model Results

- Best accuracy (92.3%): Logistic Regression, Cubic SVM, Medium NN
- Best AUC (0.972): Bagged Trees — strongest class separability
- Confusion matrices show consistently higher specificity (HC) than sensitivity (SP) across all models
- Training AUC = 1.000 for Bagged Trees/Medium NN flagged as sample memorization — cross-validated out-of-fold metrics (Bagged Trees AUC 0.972 ± 0.054 , Cubic SVM 0.959 ± 0.064) are the numbers that reflect true generalization

Model	Acc. (%)	Prec. (%)	Rec. (%)	AUC
Coarse Tree	88.2 ± 10.7	92.2 ± 11.4	84.9 ± 17.6	0.90 ± 0.10
Logistic Reg.	92.3 ± 8.7	97.3 ± 7.3	87.6 ± 15.5	0.95 ± 0.07
Naive Bayes	91.5 ± 8.7	97.2 ± 6.9	85.9 ± 15.4	0.93 ± 0.08
Cubic SVM	92.3 ± 8.2	97.1 ± 7.1	87.7 ± 14.6	0.95 ± 0.06
Fine KNN	90.0 ± 8.7	98.4 ± 5.8	81.5 ± 16.2	0.92 ± 0.08
Bagged Trees	91.6 ± 8.2	95.8 ± 8.7	88.0 ± 13.2	0.97 ± 0.05
Medium NN	92.3 ± 8.7	96.7 ± 7.7	88.0 ± 14.7	0.95 ± 0.07

Conversion of all SHAP Features to Original Features

- **Initial Feature Calculation:**

SHAP values were first computed for the individual entropy metrics (e.g., Pelvis_ApEn, Hip_SampEn).

- **Joint-Level Merging:**

Features were grouped by their anatomical joint (Ankle, Knee, Hip, Pelvis) and their respective SHAP values were summed together.

- **The Justification (Additive Property):**

Why this works: SHAP values are inherently additive. Summing the individual feature impacts accurately represents the total contribution of the entire joint.

- **Mathematical Proof:**

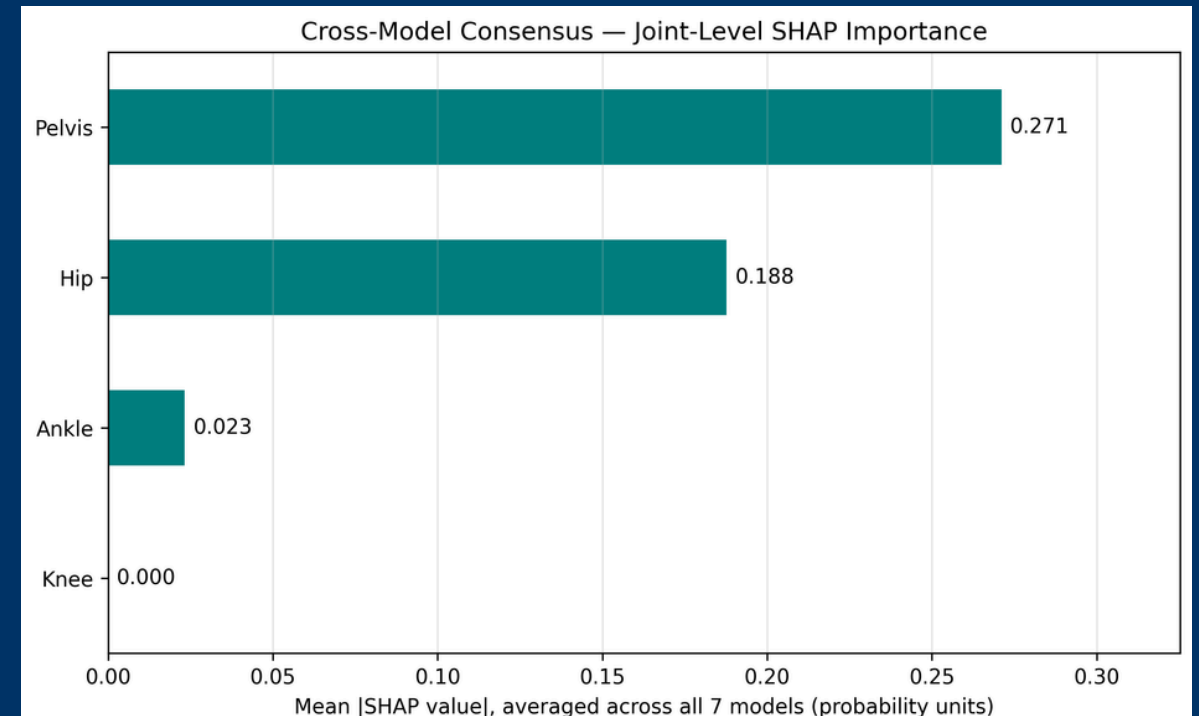
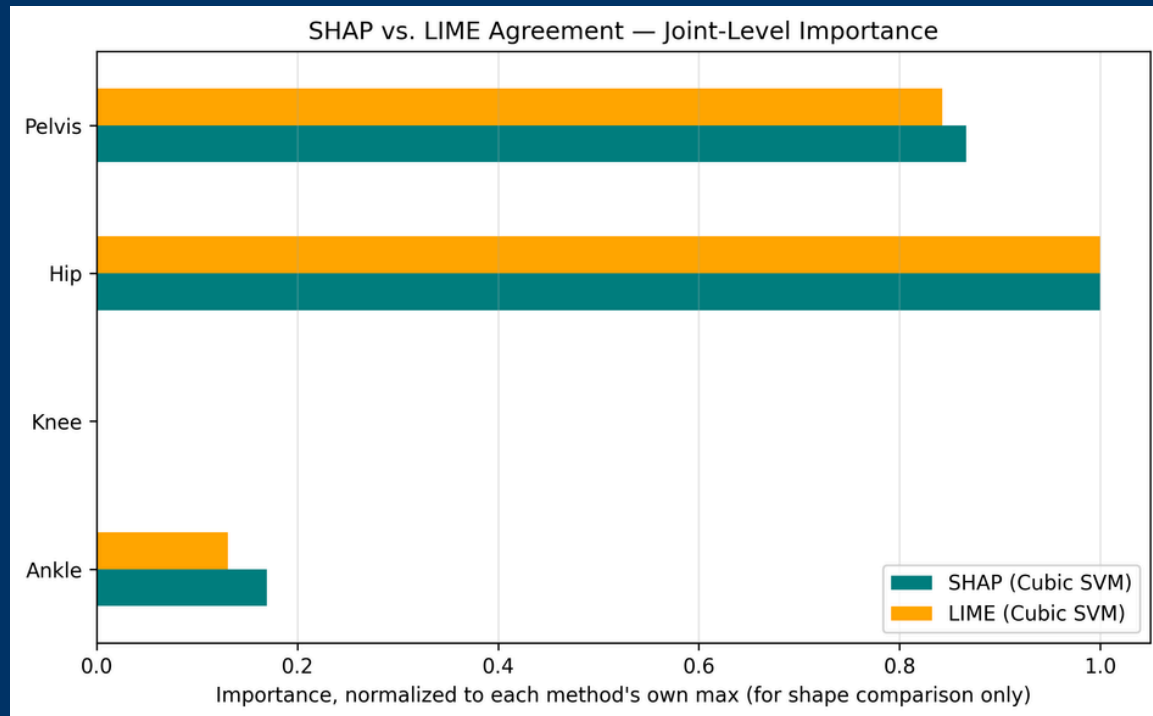
We successfully verified that no information was lost during merging. The grouped joint impacts perfectly reconstruct the model's exact prediction:

Proof: Base Value + $\Sigma(\text{Joint SHAP}) = \text{Model Output } f(x)$

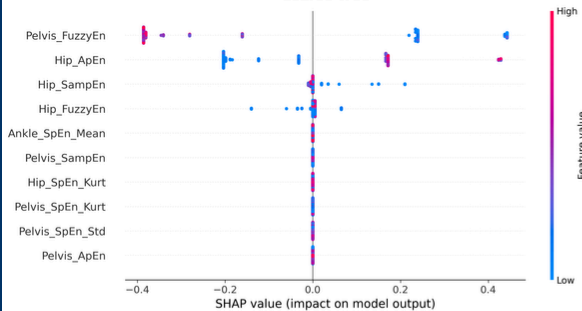
SHAP Results

Biomechanical interpretation:

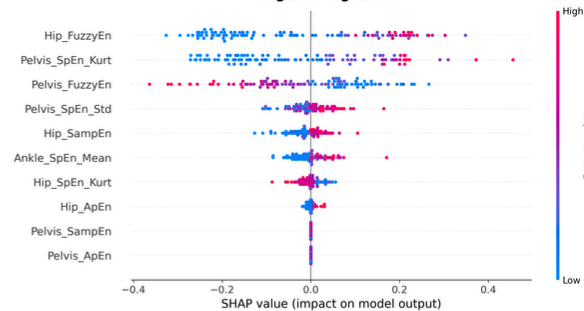
- Pelvis rigidity — lower pelvic entropy pushes predictions toward stroke (loss of movement complexity, stereotyped/rigid pelvic stabilization)
- Hip compensation — higher hip entropy pushes toward stroke (extra hip movement compensating for a weak leg, to avoid foot-scraping)
- Knee is always having value 0 means kinematics ML results almost never depend on knee angle.



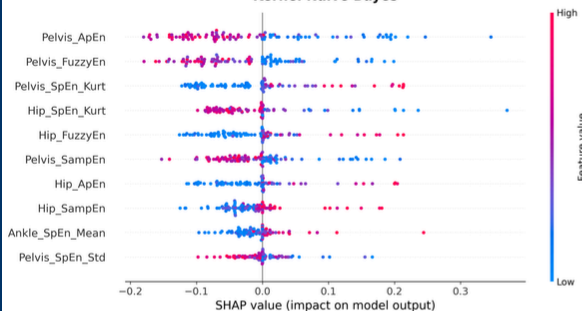
Coarse Tree



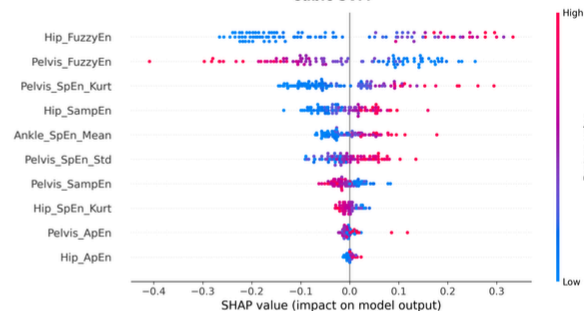
Logistic Regression



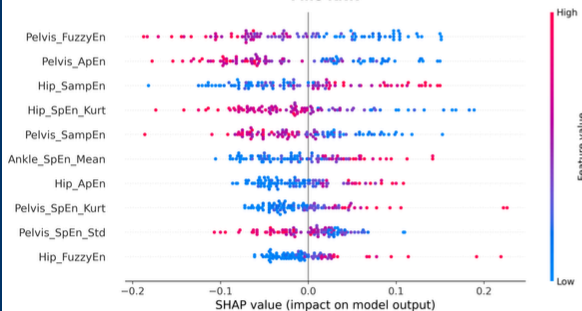
Kernel Naïve Bayes



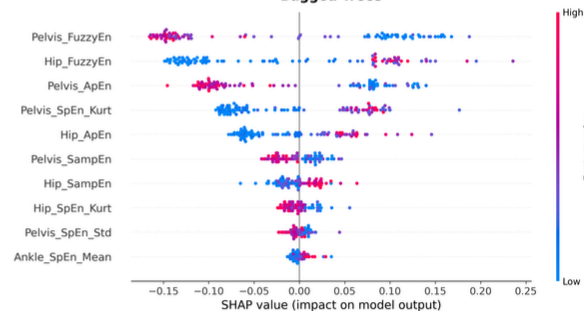
Cubic SVM



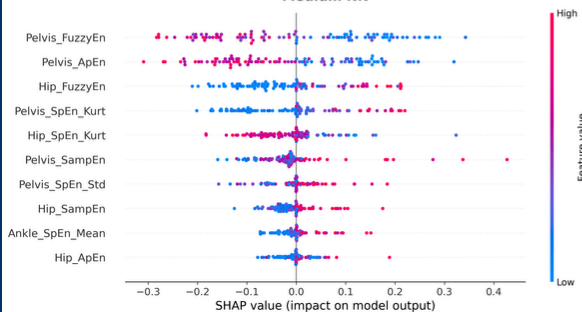
Fine KNN



Bagged Trees

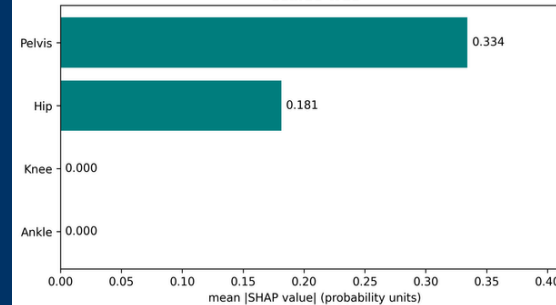


Medium NN

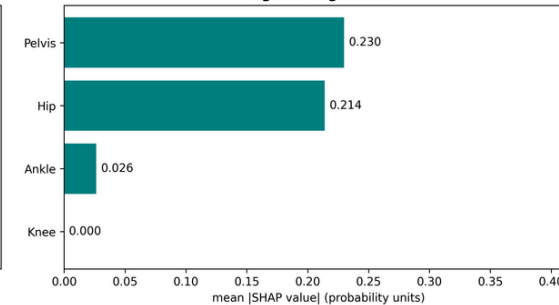


Joint-Level SHAP Importance — All 7 Models (shared scale)

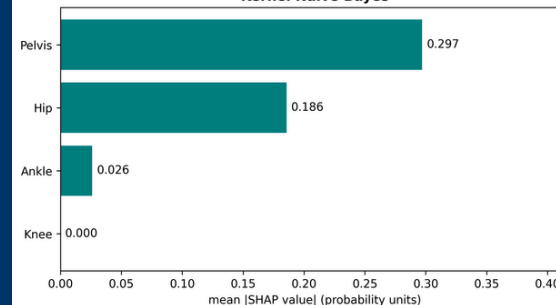
Coarse Tree



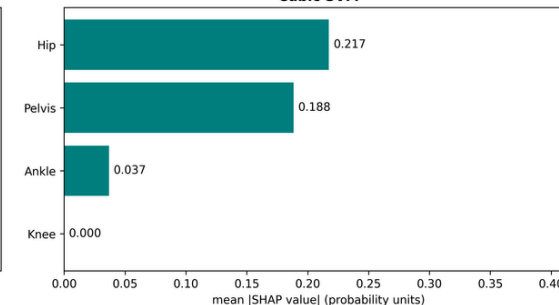
Logistic Regression



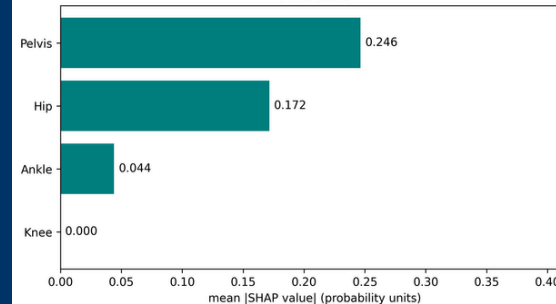
Kernel Naïve Bayes



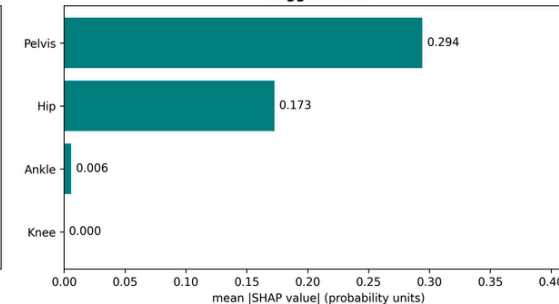
Cubic SVM



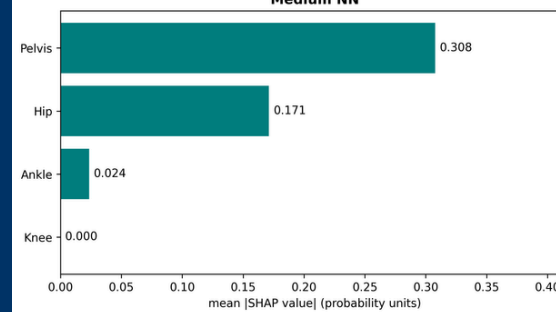
Fine KNN



Bagged Trees



Medium NN



Interactive Dashboard

- Built an interactive dashboard (Scilab GUI) that turns the trained models and their SHAP explanations into something a viewer can explore live, not just static plots
- Side-by-side 3D animated skeletons — Healthy Person vs. Stroke Patient — reconstructed from real joint-angle kinematics (pelvis, hip, knee, ankle) via forward kinematics, animated across the full gait cycle
- Synchronized 2D joint-angle graphs for Ankle, Knee, Hip, and Pelvis, with live markers tracking the same point in the gait cycle as the 3D animation — blue = healthy, black/red = stroke
- Model selector — switch live between all 7 trained classifiers (Coarse Tree, Logistic Regression, Naive Bayes, Cubic SVM, Fine KNN, Bagged Trees, Medium NN), each displaying its own cross-validated accuracy
- ML Diagnostics panel — per-joint anomaly counts computed against an adjustable sensitivity threshold, plus a live "XAI Driver" readout showing which joint's SHAP attribution is currently dominant for the selected model
- Real-time diagnosis output — Healthy Gait / Mild Deviation / Stroke Pattern, color-coded, with confidence tied to the selected model's own accuracy
- Gait-cycle scrubber + Play/Stop control — auto-animates the full cycle, or step through manually
- Optional guided-audio walkthrough button for narration

WrapUp

Key Findings & Clinical Value

- High Performance, Lower Cost: Kinematics-only entropy features match multimodal methods (92.3% Accuracy, $AUC \geq 0.95$), offering a practical, low-cost clinical monitoring tool.
- Validated Biomechanics: SHAP and LIME concordance confirms models learned true physiological mechanisms (pelvis/hip compensation) rather than spurious correlations, establishing clinical trust.

Study Limitations

- Label Granularity: Stroke group (N=50) mixes distinct presentations (e.g., foot drop, stiff knee) into a single binary label.
- Confounding Variables: Wide healthy-control age range (21–86 years) makes it difficult to fully isolate stroke effects from natural aging.
- Excluded Modalities: EMG data was excluded entirely due to raw format usability constraints in this dataset.

Future Work

- Continuous Prediction: Shift from binary classification to predicting continuous functional scores (FAC, gait speed).
- Neuromuscular Mapping: Integrate multi-channel raw EMG data for deeper activation analysis.
- Broader Validation: Test on multi-center datasets stratified by specific lesion locations and chronicity.

Thankyou :)